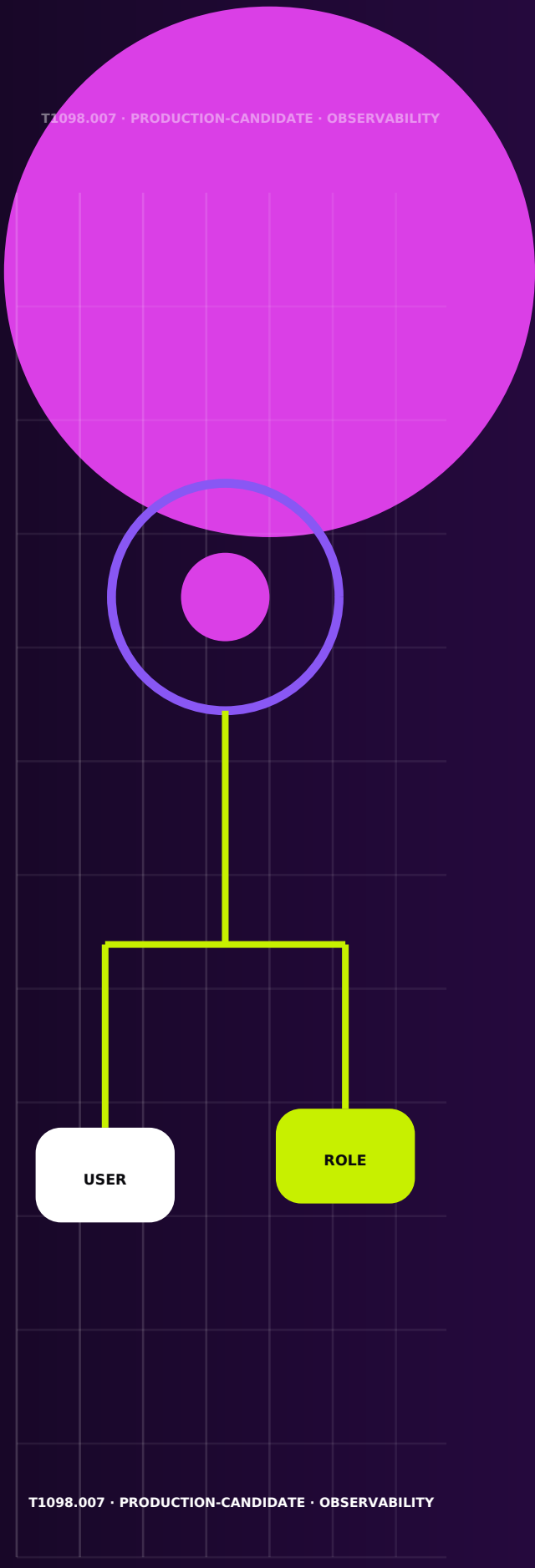
DETECTION THREAT LANDSCAPE

PAPERCUT AGENTIC CAMPAIGN - DOMAIN ADMINS MEMBERSHIP ADDITION

Detects an account being added to the Domain Admins global group by filtering its stable RID 512. Every result requires approved-change validation.



PRODUCTION CANDIDATE

CONFIDENCE: MEDIUM

REPORT TYPE
campaign

STATUS
Untested implementation sketch

PRIMARY PLATFORM
Microsoft Sentinel / Log Analytics

OPERATIONAL VALUE
10 / 10

01

Executive Assessment

Decision-grade summary with explicit evidence boundaries.

ASSESSMENT

DECISION

Retain this package as a production candidate and migrate it to the canonical visual and evidence contract without changing query conditions. Treat every result as an investigative lead until local schema, field population, benign overlap and the legacy claim provenance are reviewed.

LEGACY EVIDENCE RECORD

- The retained legacy source ledger contains 6 normalized references; claim-to-source mapping remains...

ANALYST ASSESSMENT

- The existing logic is retained at production-candidate maturity; repository migration is not...
- The folder and query header map the package to T1098.007; operational relevance still depend...

DETECTION ASSUMPTIONS

- The named telemetry is collected across the intended asset or identity scope with adequate...
- Projected fields preserve their expected semantics in the target connector and parser. [C002]
- Approved activity, owners and change records are available to analysts during disposition. [C003]

KEY FINDINGS

FINDING 1

The original query conditions and source ledger were preserved during layout migration; legacy PDF SHA-256 4f1913595179e7c4... is recorded for traceability. [C001]

FINDING 2

The principal observable is Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected investigation context. [C002]

FINDING 3

The evidence boundary remains explicit: a match does not prove successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without corroborating evidence. [C003]

EVIDENCE BOUNDARY

A matching result does not prove successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without corroborating evidence. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

02

Threat Mechanics

Technical mechanism, observable transitions and supported scope.

ATTACK FLOW

TECHNICAL BACKGROUND

The retained dossier links Papercut Agentic Campaign to the detection opportunity described as domain admins membership addition. The v2 migration preserves the query exactly and exposes its match boundary, but does not reinterpret legacy source claims as laboratory evidence. Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected investigation context. [C001][C002]

1

THREAT CONTEXT

Detects an account being added to the Domain Admins global group by filtering its stable RID 512. Every result requires approved-change validation. Observable: Context retained from the legacy dossier and source ledger.

2

OBSERVABLE CANDIDATE

Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected investigation context. Observable: Events from SecurityEvent and projected fields.

3

CONTEXTUAL ENRICHMENT

Resolve asset, identity, process, ownership and change context before disposition. Observable: Raw event, surrounding timeline, inventory and authorized-change evidence.

4

DECISION GATE

Escalate only when corroborating evidence closes the documented non-claim boundary. Observable: Controlled validation, explainable negatives and analyst-readable evidence.

03

Telemetry & Evidence Model

Minimum data contract and known visibility gaps.

DATA CONTRACT

PRIMARY PLATFORM

Microsoft Sentinel / Log Analytics. Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected investigation context.

FIELD / TABLE	STATUS	PURPOSE
SecurityEvent	**Required**	Primary event source selected by the unchanged query logic.
Event time and entity identifiers	**Required**	Chronology, deduplication, scoping and investigation pivots.
Raw projected context	**Required**	Preserve the fields emitted by the query before normalization or tuning.
Asset, identity and change inventory	**Enrichment**	Separate authorized activity and define operational ownership.

KNOWN VISIBILITY GAPS

Unvalidated local schema Missing projected fields Incomplete retention or asset scope Mutable attacker behavior Legacy claim provenance review pending

04

Primary Detection Logic

Native query, explicit condition evidence and match meaning.

KQL

```
// Rule: [T1098.007] Papercut Agentic Campaign - Domain Admins Membership Addition
// MITRE ATT&CK: T1098.007
// Classification: Production candidate
// Status: Untested implementation sketch
// Primary platform: Microsoft Sentinel / Log Analytics
// Required telemetry: SecurityEvent
// Purpose: Detects an account being added to the Domain Admins global group by filtering its stable RID 512. Every result requires approved-change
// Logic: Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected inve
// A match proves: A result is a privileged group change; validate it against the approved change record.
// A match does not prove: successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without cor
// Expected false positives: authorized administration, approved software, testing, or benign artifact collisions.
// Coverage limits: local schema, field population, retention, collection scope, and attacker variation require validation.
// Evidence claims: C001, C002, C003
//
// PaperCut campaign - account added to Domain Admins
// Status: Production candidate - Untested implementation sketch
// Confidence: Medium
// Primary platform: Microsoft Sentinel / Log Analytics
// Required connector: Windows Security Events via AMA
// Required audit policy: Audit Security Group Management on domain controllers
// Evidence: GreyNoise observed an adversary-created account being added to Domain Admins.
// RID 512 identifies the Domain Admins global group in every AD domain.
// This query has no threshold, aggregation, sequence, or join.
// A result is a privileged group change; validate it against the approved change record.

SecurityEvent
| where EventID == 4728
| where TargetSid endswith "-512"
| project
    TimeGenerated,
    Computer,
    EventID,
    Activity,
    SubjectAccount,
    SubjectUserSid,
    SubjectLogonId,
    TargetAccount,
    TargetDomainName,
    TargetSid,
    MemberName,
    MemberSid
| order by TimeGenerated desc
```

WHAT A MATCH PROVES

A result is a privileged group change; validate it against the approved change record.

WHAT IT DOES NOT PROVE

successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without corroborating evidence.

MATURITY

Production candidate - Untested implementation sketch. Evaluate SecurityEvent for domain admins membership addition using the existing source-derived conditions while preserving projected investigation context.

05

Triage & Hunting Playbook

Evidence-preserving workflow from match to disposition.

WORKFLOW

01

Preserve the complete raw matching event, query version, timestamp, connector or sensor version and stable entity identifiers.

02

Confirm the event originated from the intended Microsoft Sentinel / Log Analytics data source and verify the semantics of every condition used by the query.

03

Resolve asset, identity, process, software and destination ownership against authoritative inventories and approved change records.

04

Build the surrounding endpoint, identity, network or appliance timeline and seek an independent evidence layer before escalation.

05

Classify benign overlap explicitly; tune only with durable ownership or provenance controls, never by suppressing the core behavior globally.

06

Escalate only when corroborating evidence establishes malicious or unauthorized activity beyond the documented match boundary.

DISPOSITION RULE

Escalate only when the evidence establishes the behavior described by the query. successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without corroborating evidence.

06

False Positives & Validation

Overlap model, controlled checks and promotion gate.

QUALITY GATE

AUTHORIZED ADMINISTRATION OR SOFTWARE

Require owner, purpose, approved change and expected execution or configuration provenance.

BENIGN ARTIFACT OR COMMAND COLLISION

Require the full condition set plus an independent evidence layer or stable provenance.

PARSER, INVENTORY OR RETENTION AMBIGUITY

Reproduce a controlled event and reconcile raw fields, collection scope and authoritative inventory.

VALIDATION SEQUENCE

- Generate an authorized controlled equivalent of the primary observable without using live malicious infrastructure.
- Verify each query condition and projected field against retained raw telemetry.
- Run representative approved and near-match negative controls.
- Measure historical prevalence and classify every returned owner or workflow.
- Record parser, connector, retention and asset-scope dependencies with the test evidence.

SUCCESS

The controlled positive produces a complete, explainable row and representative negatives are separable through durable context.

REJECTION

Reject promotion when field semantics, collection scope, benign ownership or the core behavior cannot be established reproducibly.

PROMOTION GATE

Require claim-level source review, a retained controlled positive, representative negatives, measured prevalence and an accountable operational owner.

07

Analyst Assessment & References

Coverage statement, open questions and source record.

FINAL

CONCLUSION

This package now uses the canonical v2 layout and evidence contract while preserving the original detection.kql conditions. It remains a production candidate at medium confidence. The visual migration is complete; claim-level provenance and target-environment execution remain explicit promotion gates.

MITRE ATT&CK

T1098.007

PRIMARY AND AUTHORITATIVE REFERENCES

S001 greynoise.io - Observed activity: 2026-08-31 onward; campaign last-observation date not stated (2026-09-09)

hxxps://www[.]greynoise[.]io/blog/ai-orchestrated-campaign-against-papercut-ng-mf

S002 papercut.com - PaperCut - URGENT Security Advisory: PaperCut NG/MF Security Bulletin (2026-09-09)

hxxps://www[.]papercut[.]com/kb/Main/security-bulletin-27-aug-2026-urgent-security-advisory/

S003 Microsoft Learn - Event 4728: member added to a security-enabled global group (2026-08-27)

hxxps://learn[.]microsoft[.]com/en-us/previous-versions/windows/it-pro/windows-10/security/threat-protection/auditing/audit-security-group-management

S004 Microsoft Learn - Event 4728: member added to a security-enabled global group (2026-08-27)

hxxps://learn[.]microsoft[.]com/en-us/azure/azure-monitor/reference/tables/securityevent

S005 Microsoft Learn - Event 4728: member added to a security-enabled global group (2026-08-27)

hxxps://learn[.]microsoft[.]com/en-us/windows-server/identity/ad-ds/manage/understand-security-identifiers

S006 MITRE ATT&CK - Last modified: 2026-05-12 (2026-05-12)

hxxps://attack[.]mitre[.]org/techniques/T1098/007/

OPEN QUESTIONS

A matching result does not prove successful exploitation, malicious ownership, downstream impact, campaign attribution or complete compromise without corroborating evidence. Target-environment field population, benign prevalence, parser behavior and operational ownership have not been validated by this repository migration.

DOCUMENT CONTROL

Schema 2.0 | Layout 2.0 | Maturity production-candidate | Confidence Medium | No customer data | Evidence ledger required